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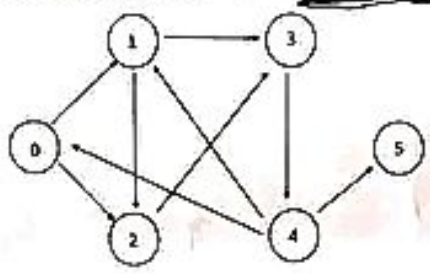
Faculty Initial: NALE
Dept. of CSE
School of Engineering
Date: Jan 31, 2026 11:30 AM

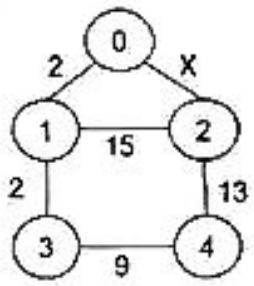
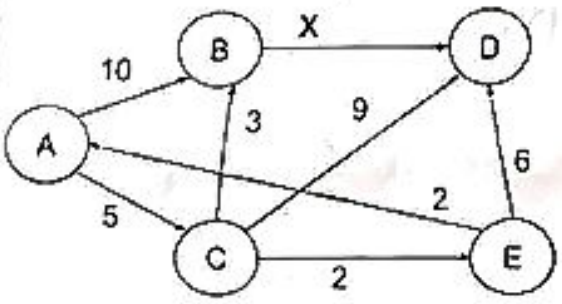
Examination	: Final Examination	**Semester**	: Fall 2025
Course Code	: CSE265	**Marks**	: 30
Course Title	: Algorithm	**Section**	: 8, 9
Slot	: Slot - 21	**Duration**	: 2 hours

General Instructions:

- All questions must be answered.
- All the devices (mobile, smartwatch, etc) should be turned off and kept in the student's bag during the exam period.

Sl. No.	Tasks	Marks	COs	BCL*
1.	Suppose, you and your friends want to make a broadcasting network for your university to spread any kind of latest news. **Determine** the appropriate algorithm for it. Write the pseudocode of the algorithm and explain the algorithm with an example. BFS	3	CO2	L-3 ...
2.	(i) **Explain** why backtracking is suitable for the N-Queens problem. (ii)Is it possible to place 3 queens on a 3x3 board? **Explain** your justification. 132 (iii) **Place** 6 queens on a 6x6 board so that no two queens attack each other. 246135	2+2+2	CO1	L-2 ...
3.	**Draw** the discovery and finishing time of each node of the following graph when DFS is applied. Source Node is **0**.  Again	2	CO3	L-3 ...
4.	**Differentiate** between BFS and DFS algorithm with examples.	2	CO1	L-2 ...
5.	Money Heist features a core group of robbers, each with a city name and specific skills, orchestrated by The Professor, a criminal mastermind who plans the heists on the Royal Mint and Bank of Spain. The professor teaches his followers to choose items technically	6	CO3	L-4 ...

*Bloom's Cognitive Level

	so that they can gain maximum profit. You can join his team if you can solve the following problem: **Items:** Item 1: Weight = 2, Value = 20 Item 2: Weight = 3, Value = 10 Item 3: Weight = 4, Value = 15 Item 4: Weight = 5, Value = 30 Item 5: Weight = 6, Value = 35 **Knapsack Capacity:** Suppose, X = last digit of your id. if (X % 2 == 0) **{Knapsack capacity, C = 12}** → 12 else if (X % 2 == 1) **{Knapsack capacity, C = 13}** **Determine** the combination of items that maximizes the total value without exceeding the knapsack's weight capacity and find the maximum profit. You cannot break items, so you must decide to take an item or leave it (0/1). 12 6 – 35 5 – 30 0 % 2 = 0 6 – 35 70 1 – 15 2 – 20			
6.	 16 Suppose, X = last digit of your id + 3 (i) **Construct** the Minimum Spanning Tree of the graph using **Prim's** algorithm. (ii)Again, **construct** the Minimum Spanning Tree of the graph using **Kruskal's** algorithm.	3+3	CO3	L-3 ...
7.	Suppose, source node is **A** and destination node is **D**. Now **calculate** the shortest path using the **Dijkstra algorithm.**  12 X = last digit of your id + 2	5	CO3	L-3 ...

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